inside the host's egg would agree in wanting the host to succeed not just in its courtship but in every detailed aspect of its life, down to being a loyal, doting parent and even grandparent.

The caddis lives inside its house, and the parasites that I have so far discussed have lived inside their hosts. The genes, then, are physically close to their extended phenotypic effects, as close as genes ordinarily are to their conventional phenotypes. But genes can act at a distance; extended phenotypes can extend a long way. One of the longest that I can think of spans a lake. Like a spider web or a caddis house, a beaver dam is among the true wonders of the world. It is not entirely clear what its Darwinian purpose is, but it certainly must have one, for the beavers expend so much time and energy to build it. The lake that it creates probably serves to protect the beaver's lodge from predators. It also provides a convenient waterway for travelling and for transporting logs. Beavers use flotation for the same reason as Canadian lumber companies use rivers and eighteenth-century coal merchants used canals. Whatever its benefits, a beaver lake is a conspicuous and characteristic feature of the landscape. It is a phenotype, no less than the beaver's teeth and tail, and it has evolved under the influence of Darwinian selection. Darwinian selection has to have genetic variation to work on. Here the choice must have been between good lakes and less good lakes. Selection favoured beaver genes that made good lakes for transporting trees, just as it favoured genes that made good teeth for felling them. Beaver lakes are extended phenotypic effects of beaver genes, and they can extend over several hundreds of yards. A long reach indeed!

Parasites, too, don't have to live inside their hosts; their genes can express themselves in hosts at a distance. Cuckoo nestlings don't live inside robins or reed-warblers; they don't suck their blood or devour their tissues, yet we have no hesitation in labelling them as parasites. Cuckoo adaptations to manipulate the behaviour of foster parents can be looked upon as extended phenotypic action at a distance by cuckoo genes.

It is easy to empathize with foster parents duped into incubating the cuckoo's eggs. Human egg collectors, too, have been fooled by the uncanny resemblance of cuckoo eggs to, say, meadow-pipit eggs or reed-warbler eggs (different races of female cuckoos specialize in different host species). What is harder to understand is the behaviour of foster parents later in the season, towards young cuckoos that are almost